

Vendor Performance Analysis

Business Problem

Effective inventory and sales management are critical for optimising profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyse the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

Data Sources

The analysis was conducted using six operational datasets obtained from the company's inventory management system:

- Beginning Inventory
- Ending Inventory
- Purchases
- Purchase Prices
- Sales
- Vendor Invoice

These datasets contain information related to inventory levels, purchasing transactions, sales performance, vendor details, pricing, and freight costs.

The datasets were integrated using Python, Pandas, SQL, and SQLite to create a consolidated analytical dataset for vendor performance evaluation.

Data Preparation

The following data preparation and transformation activities were performed:

- Imported raw CSV files into SQLite using Python and Pandas.
- Cleaned missing and inconsistent records.
- Standardized column names and data formats.
- Aggregated purchase, sales, and freight information at the vendor level.
- Calculated key business metrics such as Gross Profit, Profit Margin, and Inventory Turnover.
- Created a consolidated analytical dataset for reporting and visualization purposes.

These preprocessing steps ensured that the data was accurate, analysis-ready, and suitable for business reporting.

1. Exploratory Data Analysis

The dataset covers purchase pricing, sales volumes, freight costs, excise tax, gross profit, and stock turnover across 128 vendors and 10,692 SKUs. Records with zero or negative gross profit and zero sales quantity were excluded to focus on profitable, active inventory.

Correlation Heatmap

The heatmap below shows relationships between all key variables.

Notable findings: The heatmap reveals a strong positive relationship between purchase quantity and sales quantity, indicating that inventory procurement closely aligns with customer demand. Additionally, profit margin exhibits a weak negative relationship with sales price, suggesting that higher-priced products do not necessarily generate higher margins.

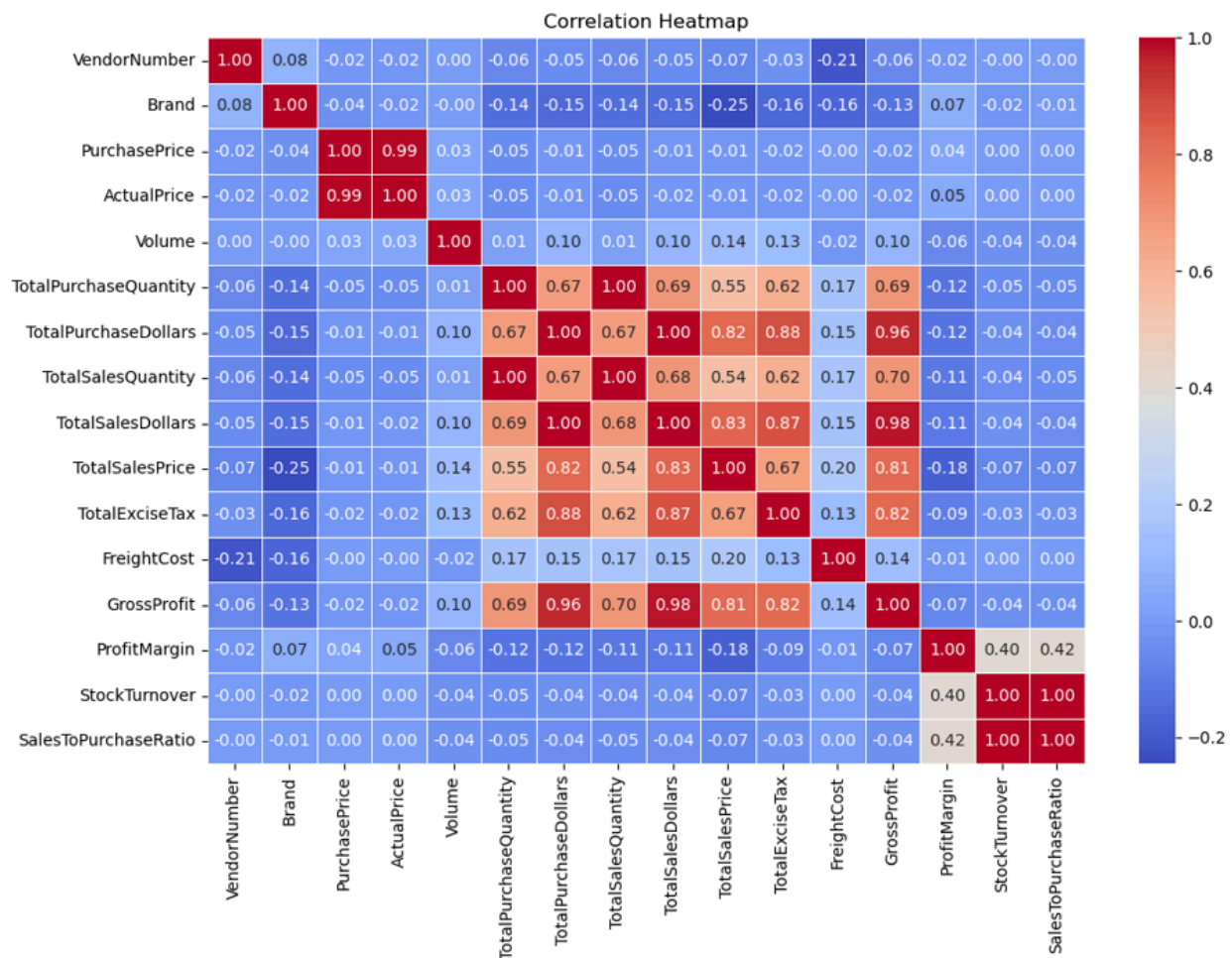


Figure 1: Correlation heatmap across all numerical variables

Data Quality Observations

- Gross profit losses: Minimum gross profit of -\$52,003, indicating items sold below purchase cost.
- Zero-sales SKUs: Some products were purchased but never sold, representing idle capital.
- Price outliers: Purchase prices range from \$0.36 to \$5,681.81; retail prices up to \$7,499.99.
- Freight cost volatility: Ranges from \$0.09 to \$257,032 across vendors and shipment sizes.
- Stock turnover range: 0 to 274.5 — some SKUs move rapidly, others are entirely stagnant.

2. Brands for Promotional or Pricing Adjustments

198 brands have been identified with below-median total sales but above-75th-percentile profit margins (>44.1%). These are strong candidates for targeted promotions or slight price reductions to unlock volume growth without sacrificing margin.

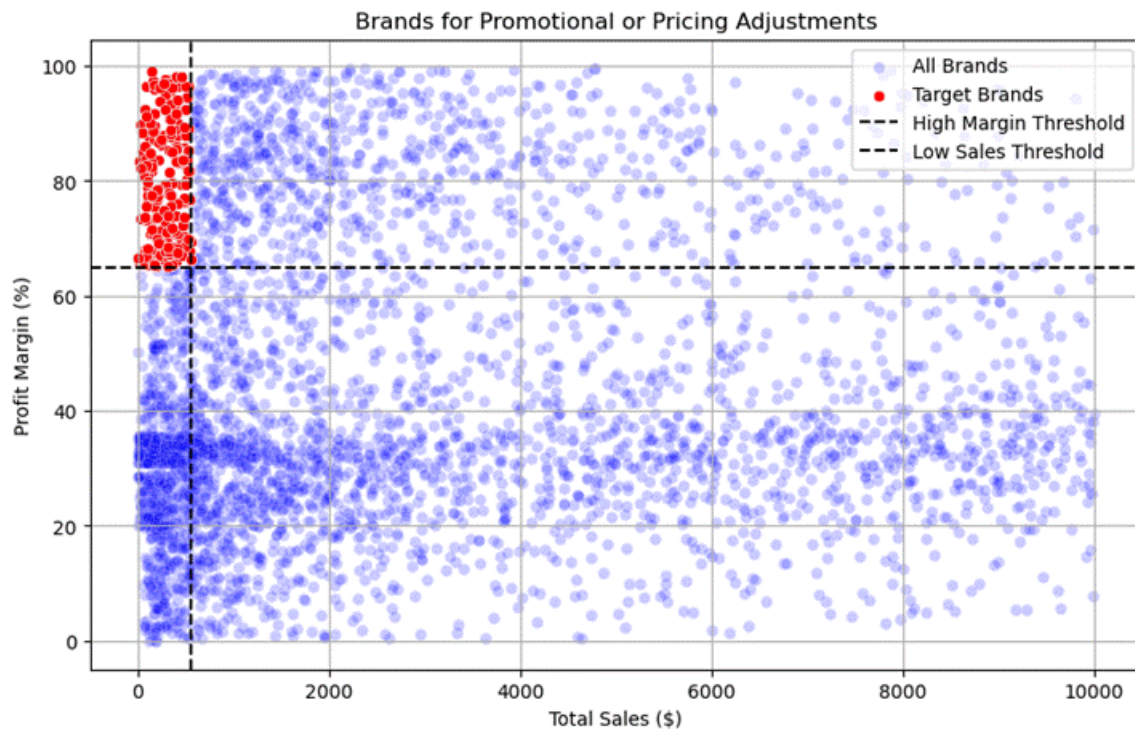


Figure 2: Brands in the top-left quadrant (red dots) have high margins but low sales volume

Sample High-Margin, Low-Sales Brands

Brand Description	Total Sales (\$)	Profit Margin (%)
Santa Rita Organic Svgn BI	\$9.99	66.5%
Debauchery Pnt Nr	\$11.58	66.0%
Concannon Glen Ellen Wh Zin	\$15.95	83.4%
Crown Royal Apple	\$27.86	89.8%
Sauza Sprklg Wild Berry Marg	\$27.96	82.2%
Dad's Hat Rye Whiskey	\$538.89	81.9%
Sbragia Home Ranch Merlot	\$549.75	66.4%
Goulee Cos d'Estournel 10	\$558.87	69.4%

3. Top Vendors by Sales & Purchase Contribution

The top 10 vendors account for 65.3% of total purchases (\$210.1M of \$322.1M). While this concentration drives efficiency, it also creates supply chain vulnerability. Vendor diversification should be a strategic priority.

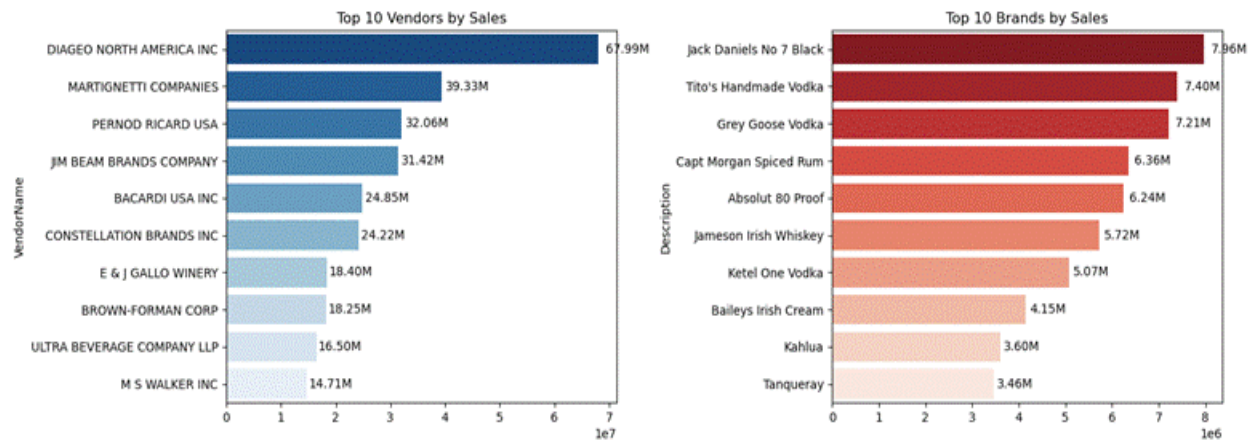


Figure 3: Top 10 vendors by total sales revenue (left) and top 10 brands by sales (right)

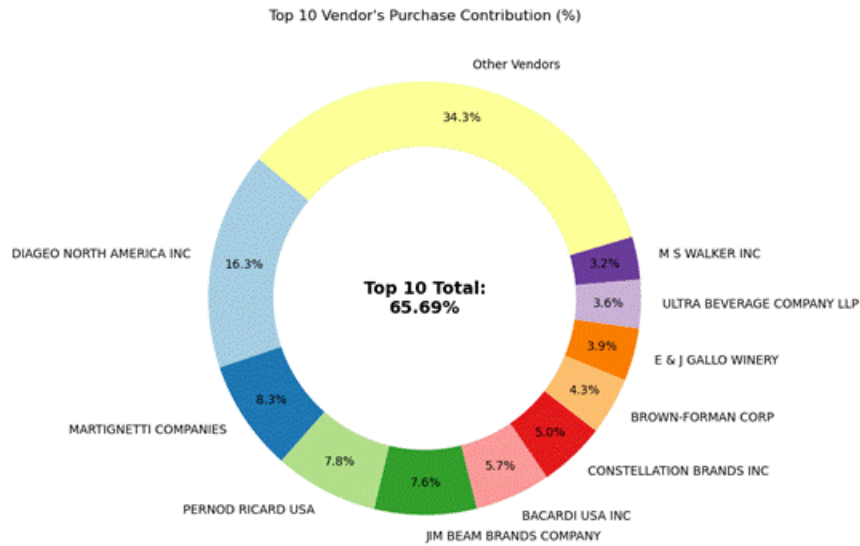


Figure 4: Top 10 vendors account for 65.7% of all purchases

Top 10 Vendors — Purchase Summary

#	Vendor	Purchase (\$)	Share	Sales (\$)	Gross Profit (\$)
1	Diageo North America Inc	\$51.0M	15.8%	\$68.7M	\$17.2M
2	Martignetti Companies	\$27.9M	8.7%	\$41.0M	\$10.8M
3	Jim Beam Brands Company	\$24.2M	7.5%	\$31.9M	\$8.4M
4	Pernod Ricard USA	\$24.1M	7.5%	\$32.3M	\$8.5M
5	Bacardi USA Inc	\$17.6M	5.5%	\$25.0M	\$6.6M
6	Constellation Brands Inc	\$15.6M	4.8%	\$24.5M	\$6.5M
7	Brown-Forman Corp	\$13.5M	4.2%	\$18.5M	\$4.9M
8	Ultra Beverage Company LLP	\$13.2M	4.1%	\$17.8M	\$4.7M
9	E & J Gallo Winery	\$12.3M	3.8%	\$18.6M	\$4.9M
10	M S Walker Inc	\$10.9M	3.4%	\$15.4M	\$4.1M

4. Impact of Bulk Purchasing on Unit Costs

Bulk purchasing delivers a 72% reduction in average unit cost — from \$40.06 (small orders under 100 units) down to \$11.39 (large orders over 1,000 units). Incentivising larger orders can significantly improve unit economics.

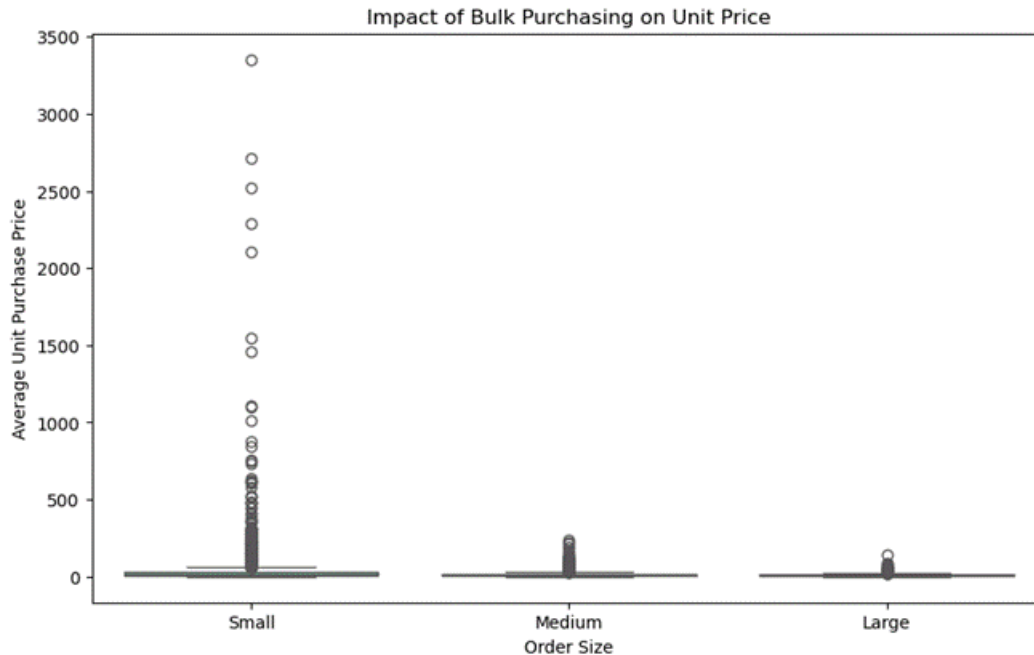


Figure 5: Distribution of unit purchase prices by order size — large orders cluster tightly at low cost

Order Size	Units Range	Avg Unit Cost (\$)	Saving vs Small
Small	< 100 units	\$40.06	—
Medium	100–1,000 units	\$17.72	56% cheaper
Large	> 1,000 units	\$11.39	72% cheaper

Recommendations

- Negotiate volume discount tiers with key vendors to formalise bulk savings.
- Introduce minimum order quantity (MOQ) incentives to push smaller buyers toward medium orders.
- Use bulk cost advantages to offer competitive retail pricing while protecting margins.

5. Inventory Turnover & Slow-Moving Stock

Total unsold inventory capital amounts to \$282,184. Slow-moving or unsold stock ties up working capital, increases warehousing costs, and risks obsolescence.

Vendors with Lowest Stock Turnover

Vendor Name	Avg Stock Turnover
Aaper Alcohol & Chemical Co	0.00
Laureate Imports Co	0.00
Truett Hurst	0.04
Vineyard Brands LLC	0.26
Highland Wine Merchants LLC	0.30
Walpole Mtn View Winery	0.82
The Imported Grape LLC	0.81
Tamworth Distilling	0.80
Smoky Quartz Distillery LLC	0.78
Centeur Imports LLC	0.77

Recommendations

- Launch clearance promotions for SKUs with zero sales in the past 90 days.
- Revise purchase orders for slow-moving vendors to reduce forward exposure.
- Consider return-to-vendor agreements for consistently unsold product lines.

6. Profit Margin: High vs Low-Performing Vendors

Statistical analysis confirms a significant difference in profit margins between top-performing vendors (mean 31.2%, 95% CI: 30.7%–31.6%) and low-volume vendors (mean 41.6%, 95% CI: 40.5%–42.6%). The null hypothesis of equal margins was rejected — these two groups operate under fundamentally different profitability models.

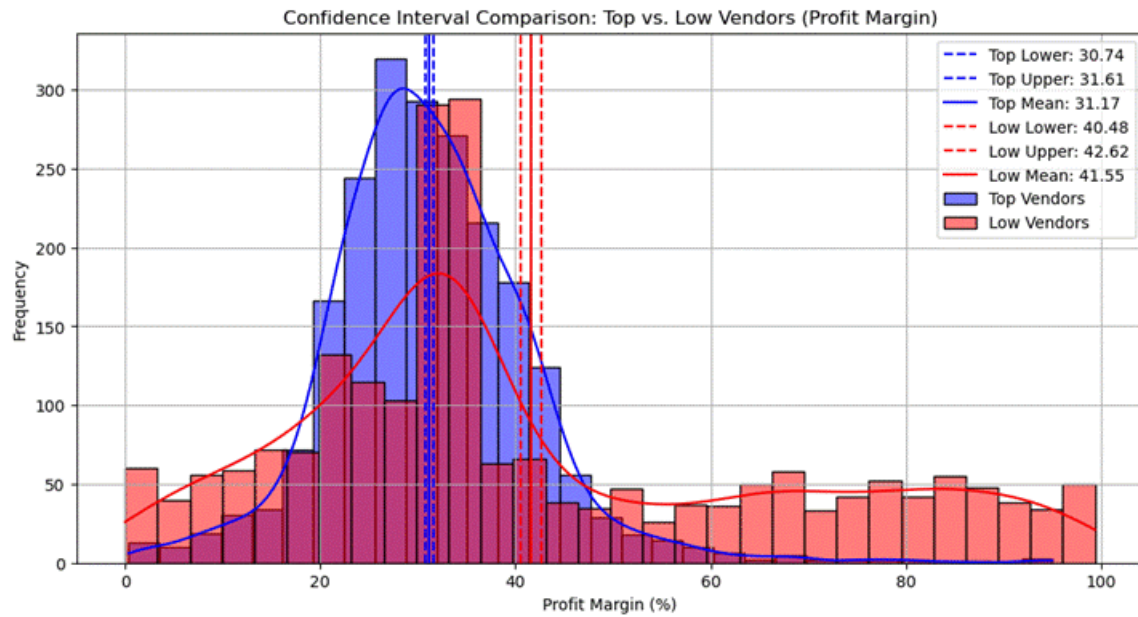


Figure 6: Top vendors (blue) cluster around 31% margin; low-volume vendors (red) spread across higher margins

Group	Mean Margin	95% Lower CI	95% Upper CI	Interpretation
Top Vendors	31.2%	30.7%	31.6%	High volume, lower margin
Low Vendors	41.6%	40.5%	42.6%	Niche pricing, low volume

Power BI Dashboard

An interactive Power BI dashboard was developed to monitor vendor performance through key metrics such as total sales, gross profit, freight costs, purchase contribution, and profit margins. The dashboard enables stakeholders to:

- Identify top-performing and underperforming vendors.
- Monitor profitability trends across vendors and brands.
- Evaluate freight costs and purchasing efficiency.
- Analyze sales performance and inventory trends.
- Support data-driven decision-making through interactive visualizations.

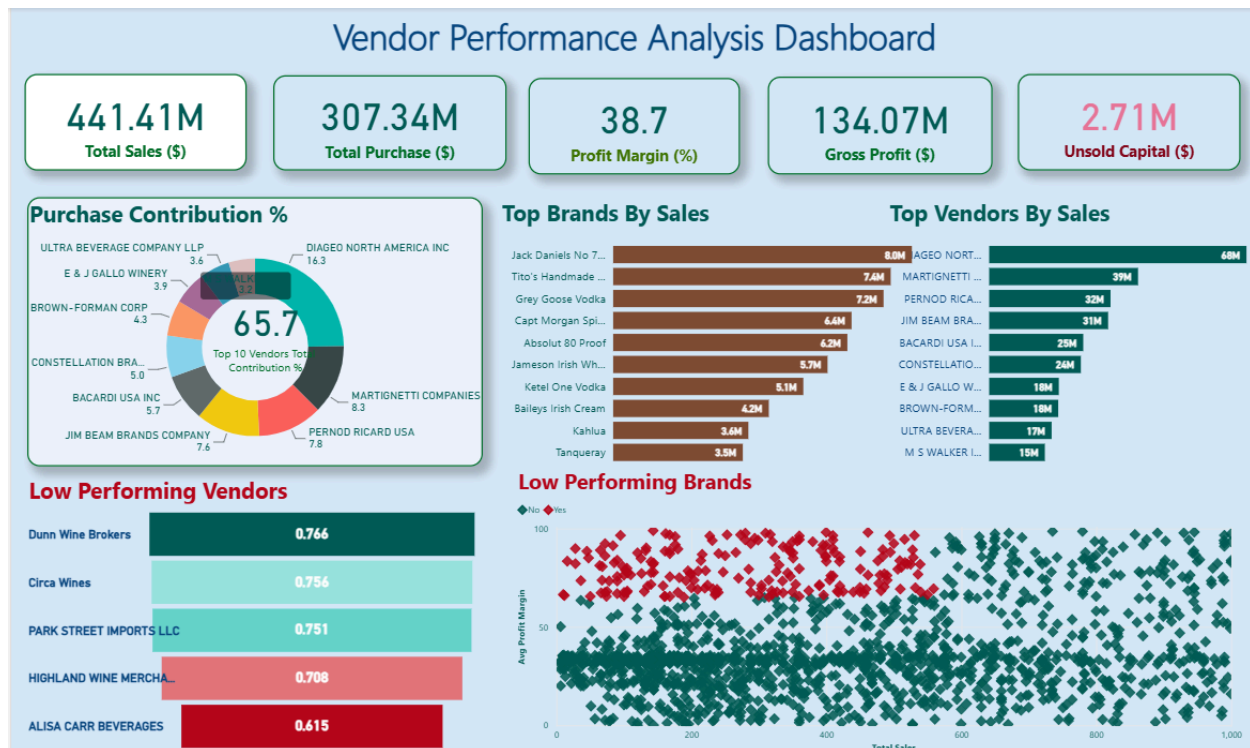


Figure 7: Interactive Power BI Dashboard for Vendor Performance Analysis

7. Final Recommendations

1. Promotional Pricing for High-Margin Brands

Re-evaluate pricing for the 198 identified high-margin, low-volume brands. A 10–15% price reduction could unlock significantly higher sales volumes without eroding profitability.

2. Vendor Diversification

The top 10 vendors control 65.3% of purchasing. Actively onboard alternative suppliers in each product category to reduce supply chain dependency and negotiate better terms.

3. Bulk Purchase Incentive Programme

Large orders deliver a 72% unit cost advantage. Formalise tiered volume discounts and pass a portion of savings to buyers to drive order size growth.

4. Slow-Moving Inventory Reduction

\$282,184 is locked in unsold stock. Implement quarterly stock reviews and clearance pricing for SKUs with zero movement. Consider vendor buy-back programmes.

5. Marketing & Distribution for Low-Performing Vendors

Low-volume vendors carry higher average margins (41.6% vs 31.2%). Investment in targeted marketing, trade promotions, and channel expansion could convert margin potential into sales volume.

6. Cost Efficiency for Top Vendors

Top-performing vendors have lower margins. Audit freight costs, excise optimisation opportunities, and operational workflows to recover 1–3 percentage points of margin.

By implementing these recommendations, the business can achieve sustainable profitability, mitigate vendor concentration risk, and improve overall operational efficiency.

Conclusion

This analysis identified several opportunities to improve profitability through vendor diversification, optimized purchasing strategies, improved inventory management, and targeted promotional activities.

The findings indicate that a small number of vendors contribute significantly to overall business performance, while certain brands exhibit high profit margins despite low sales volumes. Additionally, bulk purchasing was found to substantially reduce unit costs, highlighting opportunities for cost optimization.

By leveraging Python, SQL, SQLite, and Power BI, this project demonstrates an end-to-end analytics workflow that enables stakeholders to monitor vendor performance, improve operational efficiency, and support strategic business decision-making.